AIP | The Journal of Chemical Physics

Heterogeneous nucleation or homogeneous nucleation?

X. Y. Liu

Citation: The Journal of Chemical Physics **112**, 9949 (2000); doi: 10.1063/1.481644
View online: http://dx.doi.org/10.1063/1.481644
View Table of Contents: http://scitation.aip.org/content/aip/journal/jcp/112/22?ver=pdfcov
Published by the AIP Publishing

Articles you may be interested in

The dependence of homogeneous nucleation rate on supersaturation
J. Chem. Phys. **141**, 024307 (2014); 10.1063/1.4887338

Generalized Gibbs' approach in heterogeneous nucleation
J. Chem. Phys. **138**, 164504 (2013); 10.1063/1.4802201

Homogeneous nucleation with magic numbers: Aluminum
J. Chem. Phys. **131**, 134305 (2009); 10.1063/1.3239469

Homogeneous nucleation at high supersaturation and heterogeneous nucleation on microscopic wettable particles: A hybrid thermodynamic/density-functional theory
J. Chem. Phys. **125**, 144515 (2006); 10.1063/1.2357937

The molecular approach to heterogeneous nucleation
J. Chem. Phys. **122**, 114709 (2005); 10.1063/1.1881032

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38

Heterogeneous nucleation or homogeneous nucleation?

X. Y. Liu[a)]
Department of Physics, National University of Singapore, 10 Kent Ridge Crescent, Singapore 119260

(Received 8 October 1999; accepted 17 March 2000)

The generic heterogeneous effect of foreign particles on three dimensional nucleation was examined both theoretically and experimentally. It shows that the nucleation observed under normal conditions includes a sequence of progressive heterogeneous processes, characterized by different interfacial correlation function $f(m,x)$s. At low supersaturations, nucleation will be controlled by the process with a small interfacial correlation function $f(m,x)$, which results from a strong interaction and good structural match between the foreign bodies and the crystallizing phase. At high supersaturations, nucleation on foreign particles having a weak interaction and poor structural match with the crystallizing phase ($f(m,x)\rightarrow 1$) will govern the kinetics. This frequently leads to the false identification of homogeneous nucleation. Genuine homogeneous nucleation, which is the up-limit of heterogeneous nucleation, may not be easily achievable under gravity. In order to check these results, the prediction is confronted with nucleation experiments of some organic and inorganic crystals. The results are in excellent agreement with the theory. © *2000 American Institute of Physics.* [S0021-9606(00)51122-2]

I. INTRODUCTION

Nucleation plays an important role in almost all aspects of materials science.[1–5] The relevancy of the prediction and control of nucleation behavior is obvious across many areas of modern surface and materials science and technologies. Depending on the role of foreign bodies, nucleation can occur either via homogeneous nucleation or heterogeneous nucleation. The first corresponds to the process where the nucleation probability is homogenous thorough the system, while the latter corresponds to the process where the probability is much higher around some foreign bodies than other parts of the system.

In describing the kinetics of nucleation, many models have been published in the past 30 years.[1–4] Nevertheless, much confusion remains. The confusion is virtually associated with the understanding of the nature of nucleation, homogenous or heterogeneous nucleation. In other words, this is a question of the role which foreign bodies play in the general kinetics of nucleation.

Concerning the impact of foreign bodies on nucleation, attention was focused solely on the effect on the nucleation barrier.[6–15] The nucleation rate is given according to such classical heterogeneous nucleation theories[1,2,6–15] as

$$J=B\exp\left\{-f\frac{16\pi}{3}\frac{\Omega^2\gamma_{\rm cf}^3}{(kT)^3[\ln(1+\sigma)]^2}\right\},\quad \text{with}\ \ 0<f\leqslant 1, \tag{1}$$

where B is a kinetic constant, $\gamma_{\rm cf}$ denotes the specific interfacial free energy between the crystals and the mother phase; Ω is the volume of growth unit; and σ is supersaturation defined as $\sigma=X_A/X_A^0$ (X_A and X_A^0 represent the actual concentration and the equilibrium concentration of the solution at the given temperature, respectively).[1] In the equation, f is a parameter describing the influence of foreign particles on the nucleation barrier. In the case of homogeneous nucleation, the nucleation rate can also be described by Eq. (1) by taking $f=1$.

The implication of Eq. (1) is that heterogeneous nucleation is always kinetically more favorable than homogeneous nucleation. However, this does not explain the fact that in many cases, nucleation in different supersaturation regimes is controlled by different processes in the same system.[6] It is generally believed[1,4,7] that at low supersaturations, heterogeneous nucleation will be dominant, and at high supersaturations, homogeneous nucleation will occur. This, however, is in conflict with Eq. (1). Apart from this, the argument does not explain the following important observations (see Sec. III):

(1) nucleation can be controlled by more than two independent processes if the supersaturation changes in a much wider range;
(2) the relationship between $\ln J$ and $1/[\ln(1+\sigma)]^2$ is nonlinear for some systems at low supersaturations (c.f. Fig. 7);
(3) the interfacial tension measured under microgravity is higher than that under gravity.

Associated with these questions, another question to be addressed is whether or not we have genuine homogeneous nucleation at high supersaturations under normal conditions.

Correct answers to these questions are not only very important for the control of nucleation in general, but also very crucial for the measurement of solid–liquid interfacial tension, since nucleation experiment remains one of the most effective ways to obtain the surface free energy of crystals. By measuring the maximum supercooling of a variety of molten metal drops, Turnbull obtained from Eq. (1) ($f=1$) the interfacial free energy $\gamma_{\rm cf}\approx 0.45\Delta H_{\rm melt}/s$ for most metals, $\gamma_{\rm cf}\approx 0.32\Delta H_{\rm melt}/s$ for some metals (s is the area per

[a)]Electronic mail: phyliuxy@nus.edu.sg

 © 2000 American Institute of Physics

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38

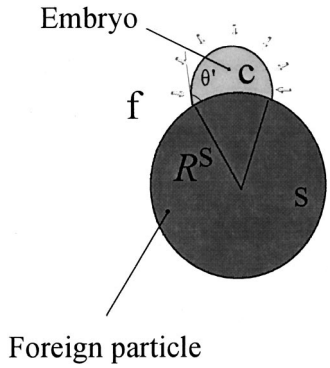

FIG. 1. Schematic illustration of heterogeneous three-dimensional nucleation on a foreign particle with a radius of R^s. Embryo: *c*, nucleating particle: *s*; mother phase: *f*.

structural unit at the surface).[16] However, later experiments indicated that much larger supercooling can be reached, suggesting that the genuine interfacial free energy could be more than 3/2–2 times higher than Turbull's estimations.[17] These results convey that the experiments carried out by Turnbull are not genuine homogeneous supersaturation. Evidently, if the effect of foreign particles cannot be eliminated under normal conditions, the surface free energy may not be measured accurately. Actually, this has been a standing issue for several decades.

In reality, it is almost impossible in most cases of crystallization to remove foreign bodies completely from crystallising systems. These foreign bodies range from bubbles, dust particles, the wall or interfaces of nucleating systems, to macromolecules and aggregates. It is the purpose of this paper to develop a new understanding on the comprehensive effect of foreign bodies on nucleation kinetics, and examine the roles of supersaturation, particle size, and interfacial properties in the kinetics of nucleation. To check the model, the predicted nucleation behavior will be confronted with experiments performed under different conditions.

II. KINETIC MODEL AND IMPLICATIONS

A. Nucleation rate under the influence of foreign bodies

In this section, we will first introduce a new kinetic model capable of describing a wide spectrum of nucleation phenomena. The molecular process of nucleation can be regarded as follows:[1] the constituent atoms or molecules in the ambient phase may on collision join into groups of two or more particles forming dimers, trimers, tetramers, etc. Before the embryos can reach a critical radius r_c they are unstable even when a positive thermodynamic driving force $\Delta\mu$ is applied. To reach r_c, an energy barrier, the so-called nucleation barrier needs to be overcome. At this stage, how the embryos reach the critical radius is the main issue. After surpassing the nucleation barrier, the second stage of the phase transition begins: growth.

We will look at the nucleation barrier for a system having foreign particles of a spherical shape with the average radius of R^s (see Fig. 1) assuming that the density of the foreign particles is N^0, and the interfacial free energies of the particle–fluid interface and of particle–crystal interface are γ_{af}, and γ_{ac}, respectively. In the process of nucleation, an embryo in increasing its size should overcome a free energy barrier, a so-called nucleation barrier ΔG^*, at a given $\Delta\mu$ ($\Delta\mu$ is the chemical potential difference between the actual state and the equilibrium state) before it develops as a stable crystal in the system. Following a similar approach proposed by Fletcher,[9] we have the free energy of formation of the critical embryo on the nucleating particle at a given supersaturation as

$$\Delta G^* = \Delta G^*_{\mathrm{homo}} f(m,x) \tag{2}$$

with

$$\Delta G^*_{\mathrm{homo}} = \frac{16\pi}{3}\frac{\Omega^2\gamma_{\mathrm{cf}}^3}{\Delta\mu^2}, \tag{3}$$

$$f(m,x) = 1/2 + 1/2\left(\frac{1-mx}{w}\right)^3 + 1/2x^3\left[2 - 3\left(\frac{x-m}{w}\right) + \left(\frac{x-m}{w}\right)^3\right] + \frac{3}{2}mx^2\left(\frac{x-m}{w} - 1\right), \tag{4}$$

$$w = (1 + x^2 - 2xm)^{1/2}, \tag{5}$$

$$x = R^s/r_c \tag{6}$$

and the critical size of nuclei

$$r_c = 2\Omega\gamma_{\mathrm{cf}}/kT\ln(1+\sigma). \tag{7}$$

Here γ_{cf} denotes the specific interfacial free energy between the crystal and the mother phase; Ω is the volume of growth units; $\Delta G^*_{\mathrm{homo}}$ corresponds to the free energy barrier of homogeneous nucleation given by Eq. (2); m depends on the interaction and structural match between the crystalline phase and the nucleating particles and is related to the interfacial tension between two different phases

$$m = (\gamma_{\mathrm{af}} - \gamma_{\mathrm{ac}})/\gamma_{\mathrm{cf}}. \tag{8}$$

(γ_{cf} corresponds to the interfacial free between the crystal and the fluid.)

The factor $f(m,x)$ varies between 1 and 0, depending on the interfacial interaction parameter m and the relative size of foreign particles x. This factor describes quantitatively the reduction of the nucleation barrier from $\Delta G^*_{\mathrm{homo}}$ to ΔG^* due to the occurrence of foreign bodies. When $x \to 0$ or $m = -1, f(m,x) = 1$. This is the case where homogeneous nucleation takes place. In the case of $m \to 1$, foreign particles reduce to embryos of the crystallizing phase. When $x \geqslant 1$, $f(m,x)$ or ΔG^* vanishes completely. This case corresponds to the growth of crystals. In the case of $x \to \infty$, a round nucleating particle reduces to a planar substrate. Then the expression (1) describes the nucleation on a plane surface. Since the value of $f(m,x)$ reflects the direct correlation between the nucleating phase and substrates in the formation of critical nuclei, we define it hereafter as *the interfacial correlation function* of heterogeneous nucleation. Under such a definition, substrates exert the strongest influence on the nucleation behavior in a relatively low supersaturation regime if the substrates and the nucleating phase is completely correlated ($f(m,x) = 0$). One the other hand, if the substrates and

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38

the nucleating phase is completely uncorrelated ($f(m,x)=1$), substrates have no influence on the nucleation behavior.

The nucleation rate will be treated based on the ''steady state''approach,[1,2,4,6,18] and the effect of foreign bodies on the dynamic of transporting growth units. Under the steady state approach, the chain process occurring in different sizes of embryos on a substrate is supposed to reach a steady state, so that the ''chain reaction'' is approximated by a quasi-''dynamic equilibrium'' state.[1,2,4,6,18] As a key point in this model, the effect of the size, the interfacial correlation function of foreign particles on the nucleation barrier, the ''chain reaction'' process, and the transport of structural units is taken into account for the nucleation taking place on these foreign bodies. The density or density distribution of foreign bodies N^0 will be considered as a parameter to be included in the final formalism.

To derive the effect of foreign bodies on nucleation rate, the distribution function describing the concentrations of embryos of various sizes should be obtained on the substrate surface.

Let ΔG_g denote the free energy barrier to form a g-mer. One then has

$$n_g \cong n_1 \exp(-\Delta G_g/kT), \tag{9}$$

where n_g and n_1 denote the density of g-mer and of monomer, respectively. Equation (9) is the expression usually quoted for the distribution of embryo sizes in supersaturated media.

Let α be the rate of molecule addition. That is,

$$\alpha = \beta'_{\mathrm{st}} \nu^{\neq}, \tag{10}$$

where $\nu^{\neq}$ is the collision rate of monomers with an embryo, and β'_{st} the sticking probability which is defined as

$$\beta'_{\mathrm{st}} = \nu \frac{a}{\lambda_0} \exp(-\Delta G^{\neq}_{\mathrm{step}}/kT) \tag{11}$$

(ν denotes the vibration frequency of structural units in the neighborhood of the surface, a is the dimension of structural units in the direction parallel to the crystal surface, λ_0 is the average distance between two kinks at the surface, and $\Delta G^{\neq}_{\mathrm{step}}$ is the activation free energy for step integration). Also let α' be the rate at which embryos lose molecules. Rates α and α' should be equal if the system were in equilibrium since no embryo would then experience a net growth or disintegration. For an embryo of radius r, the value of α is determined by the surface concentration of monomers n_1. Thus, at the detailed balance between the growth and disintegration of embryos, one has

$$\alpha_g n_g - \alpha'_{g+1} n_{g+1} = 0. \tag{12}$$

On the other hand, if there is growth, then the equilibrium distribution is perturbed. Let f_g be this perturbed surface concentration of g-mers ($f_g \leqslant n_g$). Assuming a steady state growth process J', the formation rate of critical nuclei per unit volume–time around a nucleating particle is equal to the steady state growth of embryos at the surface. This can then be expressed

$$J' = \text{critical sized nuclei formed/unit volume–time}$$

or

$$\equiv \cdots \equiv \alpha_g f_g - \alpha'_{g+1} f_{g+1} = \text{constant}. \tag{13}$$

To analyze the kinetics, the following two boundary conditions should be introduced:

$$(1) \quad f_{g_c} = 0; \tag{14}$$

$$(2) \quad \lim_{g \to 1} (f_g/n_g) = 1. \tag{15}$$

The first is because whenever a (g_c-1)-mer becomes a g_c-mer, it disappears from our population to begin a new stage—growth. But this event does not disturb the distribution of other embryos (mainly monomers). The second is because both n_g and f_g are large numbers when g is small. It follows that the perturbed concentration of monomers is almost equal to the equilibrium concentration.

Starting from Eqs. (9)–(15) and taking into account the effect of foreign bodies on the dynamics process of transporting structural units into the surface of embryos, we obtained an expression for J'[19] as

$$J' = f''(m,x)[f(m,x)]^{1/2} B \times \exp\left[-\frac{16\pi\gamma_{\mathrm{cf}}^3 \Omega^2}{3kT[kT\ln(1+\sigma)]^2} f(m,x)\right], \tag{16}$$

with

$$B = (n_1)^2 4\alpha\beta_{\mathrm{st}}\Omega \left(\frac{\gamma_{\mathrm{cf}}}{kT}\right)^{1/2}, \tag{17}$$

$$f''(m,x) = \frac{1+(1-xm)/w}{2}. \tag{18}$$

The average nucleation rate in the solution depends on the density and size of foreign particles occurring in the system and is given, according to $J = 4\pi a (R^s)^2 N^0 J'$, by

$$J = 4\pi a (R^s)^2 N^0 f''(m,x)[f(m,x)]^{1/2} B \times \exp\left[-\frac{16\pi\gamma_{\mathrm{cf}}^3 \Omega^2}{3kT[kT\ln(1+\sigma)]^2} f(m,x)\right]. \tag{19}$$

In the case of homogeneous nucleation, one has $f''(m,x') = f(m,x) = 1$, and $4\pi a (R^s)^2 N^0 \to 1$. Equation (19) then describes homogenous nucleation. We notice that the most fundamental difference between Eqs. (1) and (19) is that the influence of foreign bodies occurs not only in the nucleation barrier term (the exponential term), but also the pre-exponential term. We will see in the following sections that this difference will allow us to catch the comprehensive characteristics of heterogeneous nucleation.

B. Heterogenous nucleation: From low to high supersations

As mentioned before, the occurrence of foreign particles will lower the nucleation barrier, resulting in an increase in nucleation rate. This effect, characterized by $f(m,x)$, will be escalated by maximizing the interaction between the crystal phase and the substrate and/or increasing the relative size of

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38

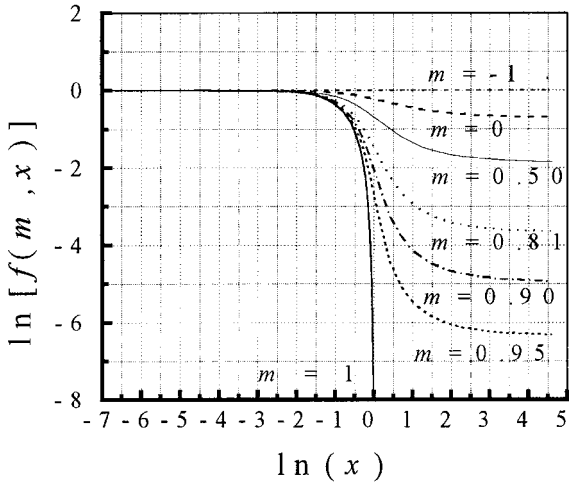

FIG. 2. Dependence of factor $f(m,x)$ on the relative particle size $x = R^s/r_c$ and the interaction parameter m between the nuncleating phase and foreign particle.

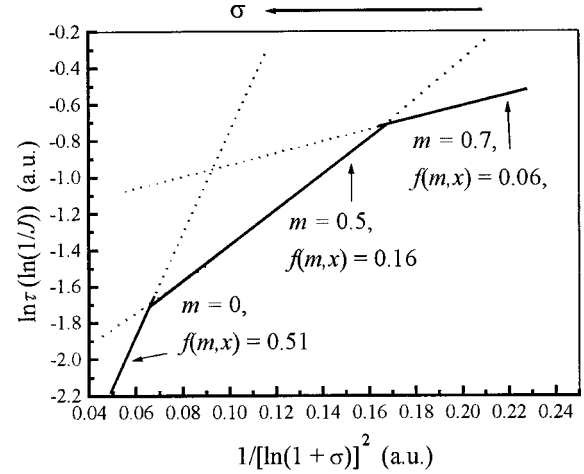

FIG. 3. The effect of interaction parameter m on the nucleation rate at different supersaturations. Foreign particles with a high m (strong interaction and better structural match with the crystallizing phase) will control the kinetics at low supersaturations while those with a low m will control the kinetics at high supersaturations. $R^s = 1000a$, $\gamma_{cf}\Omega^{2/3}/kT = 1.5$ (a is the demension of growth units).

foreign particles $x(R^s/r_c)$. The dependence of $f(m,x)$ on m and x is shown in Fig. 2. A similar plot can also be obtained for $f''(m,x)$.

On the other hand, foreign particles also exert a negative impact on the nucleation kinetics. As shown in Fig. 1, nucleation on a foreign particle will cause a reduction in the ''effective surface of embryoes,'' which is the surface of embryoes where growth units can be incorporated into the embryoes on the substrate. This tends to slow down the nucleation kinetics, which cancels the effect of the nucleation barrier lowering. This negative effect is mainly described by $f''(m,x)$ and $f(m,x)$ in the pre-exponential term of J given by Eq. (19). In this concern, the increase of m and/or x will slow the nucleation kinetics [see Eq. (19)].

In the following subsections, we will examine the effect of the size, and surface structural properties of foreign particles on the nucleation kinetics under different supersaturations. In order to compare with experimental results, we will look at the relation between the incubation of nucleation τ ($\propto 1/J$) and supersaturation. Assuming that τ is proportional to $1/J$, we should obtain a linear relation between $\ln\tau$ and $1/[\ln(1+\sigma)]^2$ for a given $f(m,x)$:

$$\ln\tau \sim \ln(1/J) = \frac{16\pi\gamma_{cf}^3\Omega^2}{3kT[kT\ln(1+\sigma)]^2} f(m,x) - \ln\{4\pi a(R^2)N^0 f''(m,,x)[f(m,x)]^{1/2}B. \tag{20}$$

1. The interaction and structural match between the crystalline phase and the substrate

In Fig. 3, $\ln\tau(\ln(1/J))$ is plotted via $1/[\ln(1+\sigma)]^2$ for different m's. As expected, for a nucleating system, foreign bodies having different surface structures and properties will control nucleation at different supersaturations. Nucleation on foreign particles with a strong interaction and good structural match between the crystalline phase and the substrate ($m\rightarrow 1$, small $f(m,x)$ and $f''(m,x)$) will be dominant at low supersaturatures, whereas nucleation on those with a week interaction and/or poor structural match between the crystalline phase and the substrate ($m\rightarrow 0$, large $f(m,x)$ and $f''(m,x)$) is kinetically favorable at high supersaturatures.

The change of nucleation process in different supersaturation regimes can be understood as follows. At low supersaturations, the nucleation barrier is very high [cf. Eq. (2)]. The top priority to accelerate the kinetics is to lower the nucleation barrier. Therefore heterogeneous nucleation will occur preferentially on the particles with a low $f(m,x)$. Conversely, at higher supersaturations, the exponential term becomes unimportant. Due to the pre-exponential factors $f(m,x)$ and $f''(m,x)$, nucleation on the substrates having a larger $f(m,x)$ and $f''(m,x)$ becomes kinetically favorable.

We notice that epitaxial growth is based on the principle of heterogeous nucleation.[1] In order to facilitate the nucleation and growth of the desired crystal phase or the specific crystallographic orietation, one can always choose the substrates with $m\rightarrow 1$. From the principle obtained in our model, we should expect that in order to obtain a better structural match, the supersaturation should not be too high, or the kinetics lead to a significant deviation of growing crystals from the desired orientation of the substrates ($m\rightarrow 0,-1$).

2. Nonlinear effect of particle size and supersaturation

$f(m,x)$ and $f''(m,x)$ are not only determined by the interaction parameter m, but also by the relative size of foreign particle x. According to Eq. (3), the relative size of foreign particle x is defined as the ratio between the radius of the particle R^s and the critical radius of nuclei. Therefore, $f(m,x)$ and $f''(m,x)$ are functions of m, R^s, and σ. The change of particles size R^s will alter x and consequently $f(m,x)$ and $f''(m,x)$. This will correspondingly change J and τ. Figure 4 shows that change of $\ln\tau$ with R^s at various supersaturations. As shown in the figure, the increase of particle size will to some extent lower $f(m,x)$ and $f''(m,x)$, and promote the nucleation rate at low supersaturations. In con-

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38

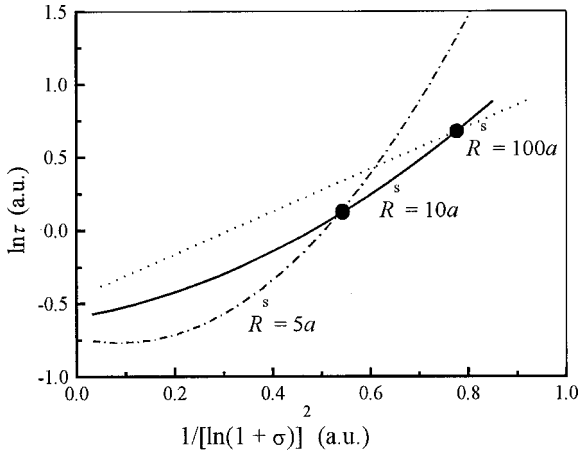

FIG. 4. The effect of particle size on the nucleation rate at different supersaturations. The effect of the size of foreign particles on the nucleation rate. Foreign particles with a large radius of curvature will control the kinetics at low supersaturations while those with a small radius of curvature will control the kinetics at high supersaturations. $m=0.8$, $\gamma_{cf}/kT=1.5$.

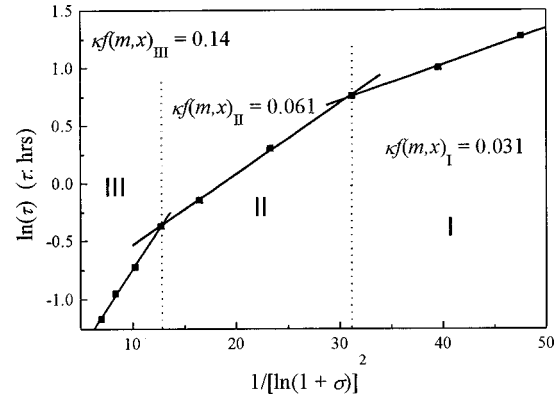

FIG. 5. Dependence of $\ln \tau$ on $1/[\ln(1+\sigma)]^2$ for paracetamol nucleating from aqueous solution. Nucleation is detected using a photodiode. The data are quoted from Ref. 20.

trast, the reduction in particles size will increase $f(m,x)$ and $f''(m,x)$ (see Fig. 2). Similar to m, it follows that small partilces will play an important role at high supersaturations.

It should be noted that due to the dependence of $f(m,x)$ and $f''(m,x)$ on the relative particles size $x(=R^s/r_c)$, which are the functions of σ, the plot of $\ln \tau \sim 1/[\ln(1+\sigma)]^2$ would not give a linear line. This nonlinearity can be clearly identified in Fig. 4. However, the lines in Fig. 3 turn out to be most linear. This is attributed to the following facts.

(1) Looking at Fig. 2, $f(m,x)$ becomes almost constant for a given m if the radius of particles R^s is much larger than the critical radius of nuclei $r_c(R^s \geqslant 50r_c)$. If the variation of r_c takes place in this regime, a linear relation between $\ln \tau$ and $1/[\ln(1+\sigma)]^2$ should be obtained. One the other hand, if the foreign particles are very small and the supersaturation is low, R^s can become comparable to the critical radius of nuclei $r_c(R^s \leqslant 30r_c)$. In this case, $f(m,x)$ will change drastically with supersaturation, and $\ln \tau \sim 1/[\ln(1+\sigma)]^2$ appears as a curve instead of a straight line (Fig. 4).

(2) The parabolic relation between r_c and σ [c.f. Eq. (7)] will have the following implications: r_c varies drastically with σ at low supersaturations, and remains a minor change with σ at high supersaturations. Therefore, we can expect the nonlinear size effect will be predominant at low supersaturations.

From the above analysis, we can conclude that in the case where the size of foreign particles has a significant effect on nucleation, a nonlinear relationship between $\ln \tau$ and $1/[\ln(1+\sigma)]^2$ should be expected. This is most likely to occur at low supersaturations where r_c is large.

In general, in a wide range of supersaturations, instead of a single process, heterogeneous nucleation will appear as a sequence of progressive processes which reveal a wide spectrum of heterogeneous characteristics described by different $f(m,x)$s and $f''(m,x)$s. Homogeneous nucleation corresponding to the case of $f(m,x)=1$ can be treated as a special case of heterogeneous nucleation. From this point of view, heterogeneous nucleation is one of the most important processes in crystallization in the whole range of supersaturations.

III. COMPARISON WITH EXPERIMENTS AND DISCUSSION

We have shown in Sec. II that foreign bodies with different $f(m,x)$ and $f''(m,x)$ will control nucleation in different regimes. According to Eq. (20) and Fig. 3, the slope of $\ln \tau \sim 1/[\ln(1+\sigma)]^2$, denoted by $\kappa f(m,x)$ ($\kappa=[16\pi\gamma_{cf}^3\Omega^2/3(kT)^3]$), should be proportional to $f(m,x)$ for a given system. A large slope corresponds to a large $f(m,x)$. Therefore, the relative $f(m,x)$ at different supersaturations can be measured by the change of the slope.

The first system which we will look at is paracetamol nucleating from aqueous solution. The nucleation was detected using a photodiode, and the measured data were published by Hendriksen and Grant.[20] The nucleation induction time of paracetamol in aqueous solution τ was plotted in the form of $\ln \tau$ vs $1/[\ln(1+\sigma)]^2$ in Fig. 5. As shown by the graph, three straight lines with different slops intercept with each other, which is similar as we predicted in Sec. II. This indicates that foreign bodies with three different $f(m,x)$s control nucleation in three different regimes, namely regimes I, II and III. From regime I to III, the supersaturation increases progressively. If we denote $f(m,x)$ and $f''(m,x)$ occurring in these three regions as $[f(m,x)]_{\mathrm{I}}$, $[f''(m,x)]_{\mathrm{I}}$; $[f(m,x)]_{\mathrm{II}}$, $[f''(m,x)]_{\mathrm{II}}$; $[f(m,x)]_{\mathrm{III}}$, $[f''(m,x)]_{\mathrm{III}}$, respectively, according to the stopes of the straight lines, we then have $[f(m,x)]_{\mathrm{I}}<[f(m,x)]_{\mathrm{II}}<[f(m,x)]_{\mathrm{III}}$ (and $[f''(m,x)]_{\mathrm{I}}<[f''(m,x)]_{\mathrm{II}}<[f''(m,x)]_{\mathrm{III}}$). This agrees completely with our predictions given in Sec. III (cf. Fig. 3).

Looking at the linearity of $\ln \tau \sim 1/[\ln(1+\sigma)]^2$ obtained in Fig. 5, we can conclude that the change of $f(m,x)$ at different supersaturations is attributed to the change in m rather than the size of foreign bodies.

Similar results were obtained for *N*-lauroyl-*L*-glutamic acid di-*n*-butylamide nucleating from iso-stearyl alcohol solutions (Fig. 6), and $CaCO_3$ from aqueous solution[6] (Fig. 7). We notice that the interfacial free energy of $CaCO_3$ γ_{cf} is normally much higher than organic crystals. It follows from

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38

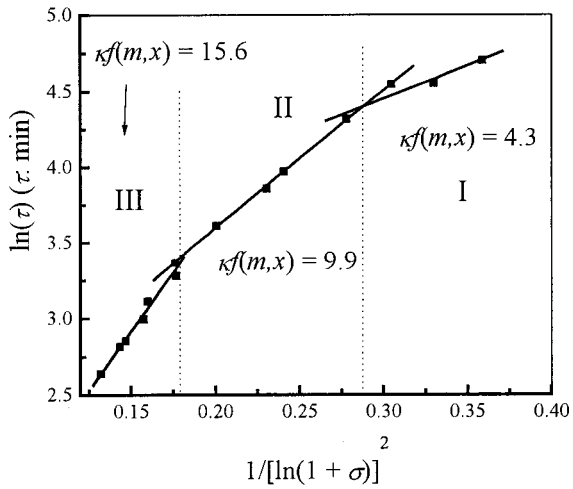

FIG. 6. Dependence of $\ln \tau$ on $1/[\ln(1+\sigma)]^2$ for *N*-lauroyl-*L*-glutamic acid di-*n*-butylamide nucleating from iso-stearyl alcohol solutions. Within the range of supersaturations where the experiments were carried out, three straight lines with different slopes intercept each other, dividing the space into three regimes. *N*-lauroyl-*L*-glutamic acid di-*n*-butylamide (>95%) and iso-stearyl alcohol (>99.0%) are obtained from Aijnomoto and Unichema, respectively. The induction time for *N*-lauroyl-*L*-glutamic acid di-*n*-butylamide nucleation was observed by the change of the gelation property of the solution. This is because *N*-lauroyl-*L*-glutamic acid di-*n*-butylamide crystals form fiber networks. Once the nuclei occur in the solution, the growth will take place rapidly. It was observed that a tiny amount of the crystals occurring in the system will gel liquid locally. From the change of the rheology property of the solutions, we can then measure the incubation time of nucleation.

Eq. (7) that at low supersaturations, the radius of critical nuclei is much bigger. According to our prediction in Sec II, the nonlinear effect of relative particle size x might take place at low supersaturations. Looking back to Fig. 7, it can indeed be identified that at low supersaturations (regime I), the experimental data are fitted much better by a curve (the dot line 2) than a straight line (the solid line 1).

In Sec II, we concluded that within the framework of our current model, genuine homogeneous nucleation can be treated as an up-limit of heterogeneous nucleation, which might occur at extremely high supersaturations. It can be noticed from the example given above that within the range of supersaturation, the plot of $\ln \tau \sim 1/[\ln(1+\sigma)]^2$ always gives the highest slope of the straight line in regime III, as expected. A crucial question to be addressed is as to whether the nucleation occurring in regime III is the genuine homogeneous nucleation. The answer is NO. According to Eq. (20), if the genuine homogeneous nucleation occurs, we should have $f(m,x)=1$, and the slope of $\ln \tau \sim 1/[\ln(1+\sigma)]^2$ will reach its uplimit. However, the slope for the nucleation of $CaCO_3$ under microgarvity in regime III is 42, about four times higher than that under gravity.[21–23] This clearly indicates that the nucleation occurring under gravity in regime III is not genuine homogeneous nucleation.

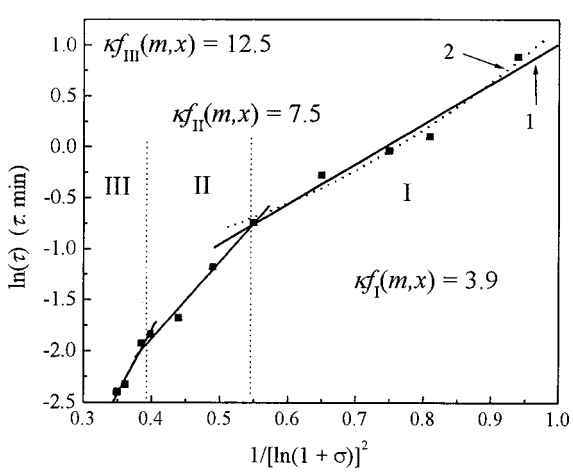

FIG. 7. $\ln \tau$ vs $1/[\ln(1+\sigma)]^2$ is plotted for the nucleation of $CaCO_3$ in gravity, three straight lines of different slopes intercept each other, dividing the space into three regimes. $\ln \tau$ vs $1/[\ln(1+\sigma)]^2$ in regime I is slightly nonlinear, due to the nonlinear effect of relative size.

In the presence of foreign bodies, to achieve genuine homogeneous nucleation may require substantially high supersaturations, which may be inaccessible. To remove these particles by filtering solutions may be one of key steps in achieving homogeneous nucleation. On the other hand, the effect of the interfaces between crystallization systems and their surroundings (such as the wall of crystallization vessels) on nucleation is impossible to be eliminated under gravity.

On the other hand, the nucleation under microgravity is much closer to genuine homogeneous nucleation. It is indicated that the interfacial free energy obtained from the microgravity experiments is very close to that obtained from the contact angle measurements ($f(m,x)\sim 1$).[23] From this point of view, microgravity will promote homogeneous nucleation.

The microgravity effect on nucleation can be understood in terms of convection. Under gravity, convection due to temperature or concentration gradient caused by nucleation and growth helps to transport growth units to the surface of substrates. This will compensate for the concentration (or supersaturation) depression during nucleation. It will also help to bring new seed crystals from substrate surface to the bulk, which virtually enhances the ''number'' of effective foreign bodies for heterogeneous nucleation. In general, convection will substantially promote the effect of foreign bodies, such as tiny particles and organic bodies or the wall of crystallization vessels, which inevitably occur in all crystallization systems. Once heterogeneous nucleation occurs at relatively low supersaturation, nucleating materials are consumed quickly. This then leads to a drop of the bulk supersaturation. Therefore, the supersaturation required for genuine homogeneous nucleation becomes extremely difficult to reach.

Under microgravity, the convection is suppressed. This was observed in the microgravity experiments.[21,22] Microgravity, therefore, will significantly slow down the transport of growth units toward the substrate surface, and cause the concentration (or supersaturation) depression during nucleation on the surface of substrates. In this case, in order to achieve a similar nucleation rate for heterogeneous nucleation, much larger supersaturation is needed. Therefore, *in microgravity, heterogeneous nucleation is suppressed to a larger extent so that homogeneous nucleation may be achieved more easily*.

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38

In summary, most nucleation observed under gravity is most likely to be heterogeneous nucleation. Within a wide range of supersaturation, heterogeneous nucleation can be a series of independent processes, which are described by $f(m,x)$. Among those, the heterogeneous nucleation on the foreign bodies has a strong interaction and good structural match with the nucleating phase will be dominant at low supersaturation while the heterogeneous nucleation on the foreign bodies has a weak interaction and poor structural match with the nucleating phase will control the kinetics at high supersaturation. The latter frequently leads to the false identification of homogeneous nucleation. Genuine homogeneous nucleation is an up-limit of heterogeneous nucleation, which may require extremely high supersaturation or a microgravity environment, which may not be easily achieved under gravity.

[1] A. A. Chernov, *Modern Crystallography III-Crystal Growth* (Springer, Berlin, 1984).
[2] B. Mutaftschiev, in *Handbook on Crystal Growth*, edited by D. T. J. Hurle (North-Holland, Amsterdam, 1993), p. 187.
[3] R. Lacmann and P. Schmidt, in *Current Topics in Materials Science*, edited by E. Kaldis and H. J. Scheel (North-Holland, Amsterdam, 1977), Vol. 2, pp. 301–325.
[4] V. P. Skripov, in *Current Topics in Materials Scierce*, edited by E. Kaldis and H. J. Scheel (North-Holland, Amsterdam, 1977), pp. 327–378.
[5] N. E. Chayen, J. Appl. Crystallogr. **30**, 108 (1997).
[6] A. I. Malkin, A. A. Chernov, and I. V. Alexeev, J. Cryst. Growth **97**, 765 (1989).
[7] O. Söhnel and J. W. Mullin, J. Cryst. Growth **44**, 377 (1978).
[8] N. H. Fletcher, J. Chem. Phys. **29**, 572 (1958).
[9] M. Volmer, *Kinelik der Phasenbildung* (Steinkipf, Dresden, 1939).
[10] R. Kaischew, Bull. Bulg. Acad. Sci. (Phys. Series) **2**, 191 (1951).
[11] L. Krastanov, Meterorol. Gidrol. SSSR (Moscow) **No:12**, (1957).
[12] R. Kaischew and B. Mutaftschiev, Bull. Chem. Inst., Bulg. Acad. Sci. **7**, 145 (1959); **3**, 5 (1963).
[13] B. Vonnegut, J. Colloid Sci. **3**, 563 (1948).
[14] R. Kaischew and B. Mutaftschiew, *Proceedings of the 1st National Conference on Aerosols* (Czechoslovakian Academy of Sciences Press, Prague, 1965), p. 293.
[15] J. Zell and B. Mutaftschiev, J. Cryst. Growth **3/4**, 535 (1968).
[16] J. Turnbull, J. Appl. Phys. **21**, 1022 (1950).
[17] T. Takahashi and W. A. Tiller, Acta Metall. **17**, 643 (1969); D. W. Gomersall, S. Y. Shiraishi, and R. G. Ward, J. Aust. Inst. Met. **10**, 220 (1965); G. A. Colligen, W. T. Loomis, and V. A. Surorenant, *ibid.* **10**, 89 (1965); G. L. F. Powell, *ibid.* **10**, 223 (1965); M. J. Stowell, Philos. Mag. **22**, 1 (1970).
[18] L. Farkas, Z. Phys. Chem. (Munich) **125**, 236 (1927).
[19] X. Y. Liu, J. Chem. Phys. **111**, 1628 (1999).
[20] B. A. Hendriksen and D. J. W. Grant, J. Cryst. Growth **156**, 252 (1995).
[21] K. Tsukamoto, Extended Abstract of AIST Workshop Sapporo, October 1998, pp. 29–33.
[22] K. Tsukamoto, S. Maruyama, K. Shimizu, H. Kawasaki, and T. S. Morita, Rep. Microgravity Experiments Aircraft **7**, 51 (1998).
[23] K. Tsukamoto and X. Y. Liu (unpublished).

This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to IP: 152.23.0.100 On: Mon, 24 Nov 2014 20:17:38